Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Team KaigoAI
b. Team leader name: Saumya Singh
c. Problem Statement: Sustainability & Social Impact

kaigoAI

Care, uninterrupted.

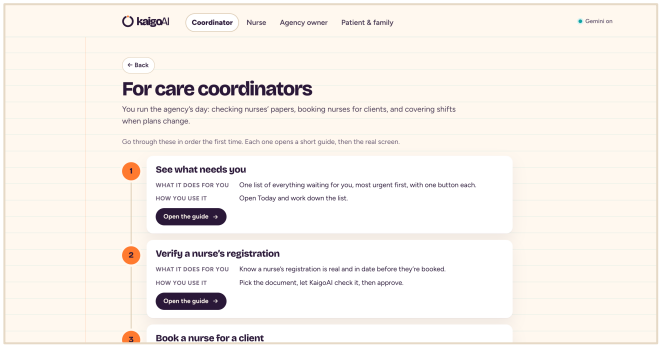
AI operations for home-care agencies in India

Team KaigoAI: Saumya Singh, Shambhavi Dubey

Brief about the idea

A nurse cancels at 21:32. Care carries on.

KaigoAI is an AI operations layer for home-care agencies in India. It checks caregivers' registration papers, turns plain-language care requests into requirements, matches the right caregiver, and replans within minutes when someone cancels.
A coordinator approves every change. Gemini reads and explains; plain rules decide. The app has one tab for each person, with a guide for every function.



Live: one tab for each person (here, the coordinator)

TRUST

Papers, checked.
Gemini reads each registration; seven rules decide what's valid.

FIT

The right person, nearby.
Requests in plain words; ranked on seven transparent factors.

CONTINUITY

Cover, before the panic.
When someone cancels: plans, a grounded explanation, approval.

Opportunities

How is it different from existing ideas?

— Staffing tools ask "who is free?". KaigoAI asks "how does this patient's care stay continuous?"

— It builds full, split and partial cover plans and scores familiarity and handovers, not just availability.

— Every value shows who produced it: Gemini, a rule, or a person.

How will it solve the problem?

— Papers checked in seconds: Gemini reads, seven rules decide, unclear cases go to a person.

— Requests become requirements; ambiguous dates are confirmed before matching.

— On a cancellation: the affected shift is found, plans are built and explained, the coordinator approves, and the new caregiver gets a booking and handover note.

USP of the proposed solution

— Care Continuity Intelligence: replanning with trade-offs, not just matching.

— Grounded AI: an explanation citing a number not in the data is withheld.

— Human-in-the-loop by design; works even when AI is down.

— One engine; new care types by configuration.

List of features offered by the solution

1 Reads credentials

Gemini turns certificate images into fields with a confidence each; unreadable values are never invented.

2 Seven validity rules

Expiry, council, number format, evidence. Blocked documents can't be approved.

3 Plain-language requests

Requirements from everyday words; ambiguous start dates must be confirmed; diagnoses are dropped.

4 Transparent matching

Seven factors incl. real driving time (Maps) and familiarity; exclusions explained by rule.

5 Replanning

Full, split or partial plans; infeasible plans can't be approved; fewer handovers wins ties.

6 Grounded explanations

Gemini explains with the rules' own numbers and drafts non-clinical handover notes.

7 Caregiver chat

Messages are understood in the app; bookings and notices go out automatically.

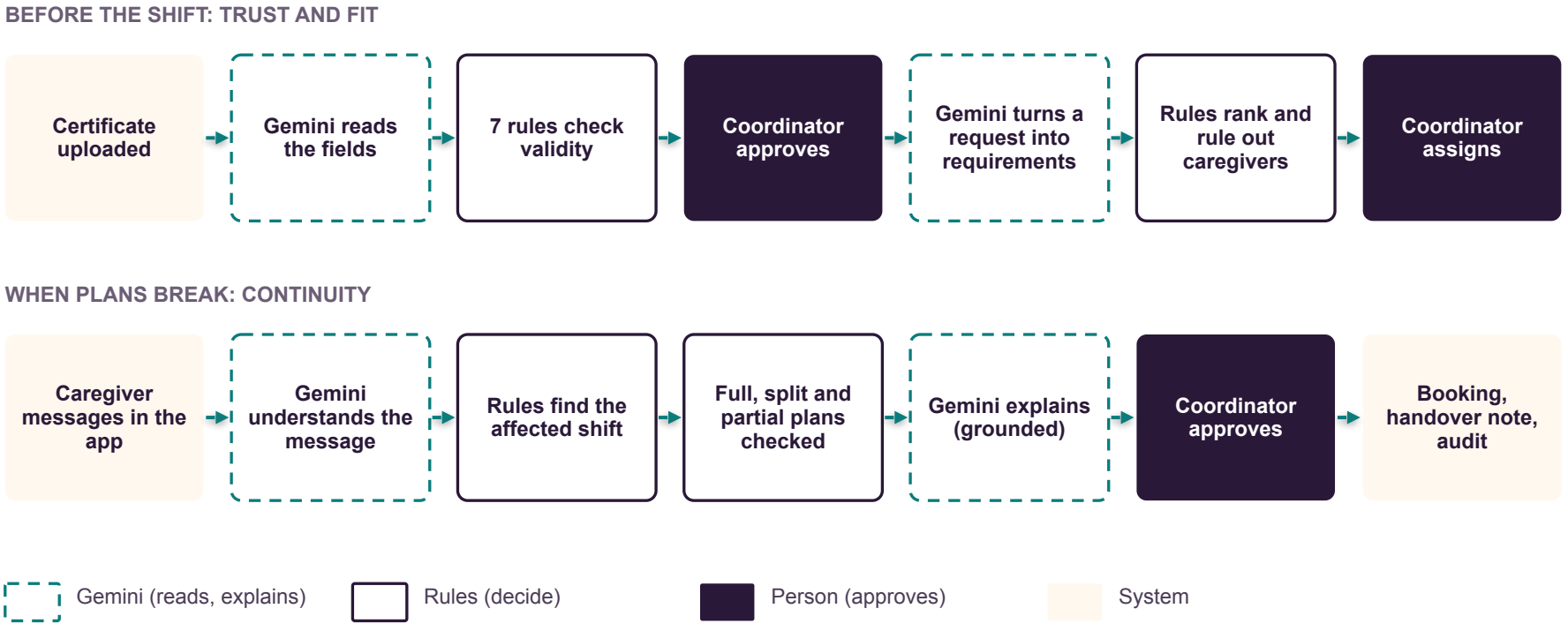
8 Proactive checks

A nightly Cloud Scheduler job finds registrations that lapse before a shift ends.

9 Approval and audit

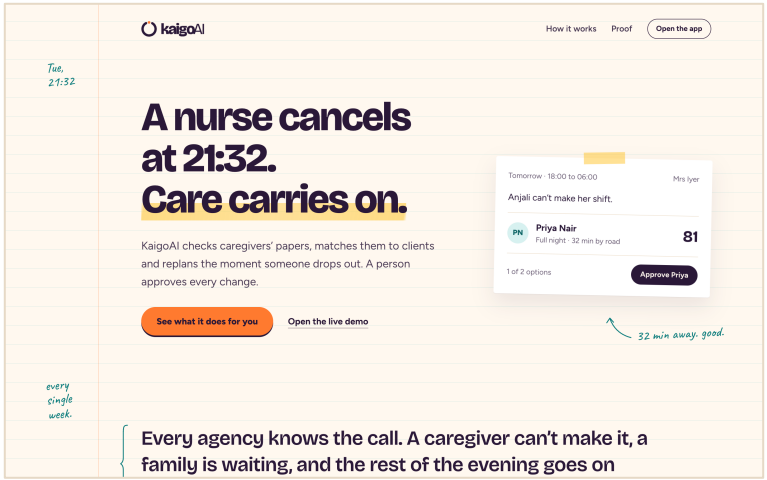
Every change waits for a coordinator; every AI, rule and human step is logged.

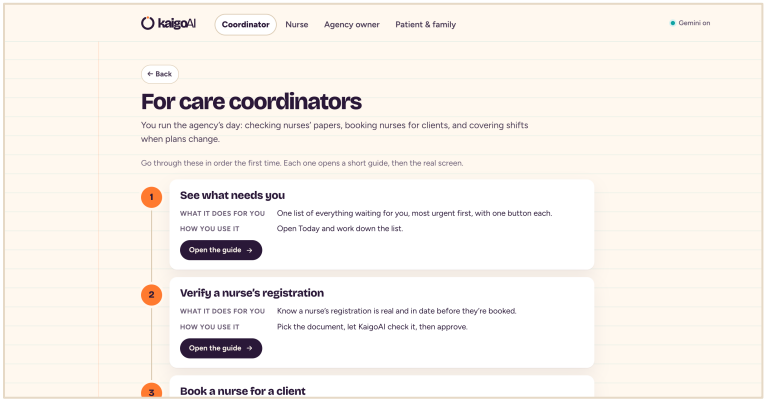
Process flow diagram or Use-case diagram

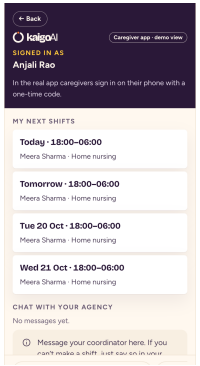


Nothing changes until a person approves; every step is written to the audit trail.

Wireframes/Mock diagrams of the proposed solution (optional)







Website

One calm page on lined paper; pen notes point at what matters; one button opens the app.

Coordinator tab

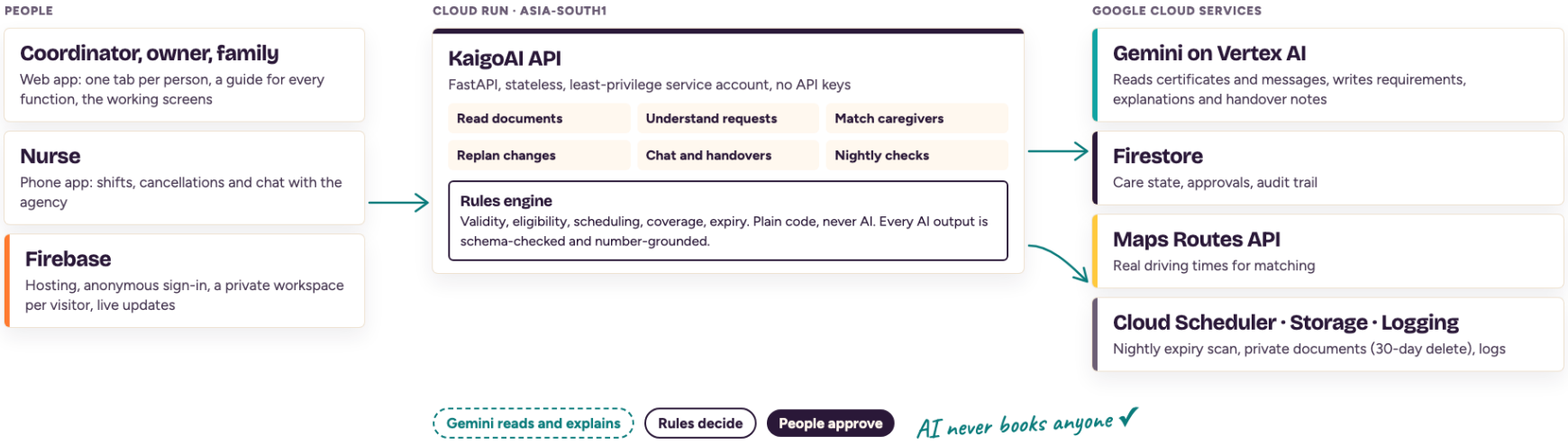
One tab per person (coordinator, nurse, owner, family); each lists what KaigoAI does for them, with a guide and Try it now.

Caregiver phone

Shifts, chat, quick replies; bookings arrive as notes.

Design system: the KaigoAI brand (logo, sunrise colours, notebook paper, pen notes), applied in the live app.

Architecture diagram of the proposed solution



Firebase serves the app and gives every visitor a private workspace. Cloud Run runs the agents and the rules engine. Gemini on Vertex AI reads documents and messages and explains trade-offs. Firestore holds the care state and the audit trail.

Technologies to be used in the solution

Gemini 3.8 Flash on Vertex AI

Reads certificates (image), requests and caregiver messages into schema-checked JSON; explains options and drafts handover notes. Called by the service account, no API keys; busy responses retried.

Cloud Run

FastAPI backend: intake, verification, matching, replanning, messaging and the rules engine. Stateless, min 1 / max 5 instances, asia-south1.

Firestore

All care state: caregivers, credentials, requests, shifts, change events, approvals, chat and the audit trail; one isolated workspace per visitor.

Firebase

Hosting for the website and app (with /api rewrites to Cloud Run), anonymous Authentication, real-time listeners for live updates.

ALSO ON GOOGLE CLOUD

Maps Routes API · Cloud Scheduler (OIDC) · Cloud Storage · Cloud Logging · IAM least-privilege service accounts

APPLICATION

Python 3.12 · FastAPI · Pydantic v2 · google-genai SDK · React 19 · TypeScript · Vite 7 · React Router 7 · Firebase JS SDK

QUALITY AND SECURITY

61 automated backend tests · 84-case UAT runner (Chrome DevTools) · ruff · gitleaks on every commit · Firestore security rules

DATA

Synthetic people and documents only; public regulatory references cited separately

Estimated implementation cost (optional)

Gemini step (measured live)	Input tokens	Output tokens	Cost 2026	Cost 2027
Read a certificate	2,618	270	$0.0030	$0.0060
Understand a care request	1,009	221	$0.0016	$0.0032
Understand a caregiver message	661	84	$0.0008	$0.0016
Explain replacement options	1,660	325	$0.0025	$0.0049
Draft a handover note	522	142	$0.0009	$0.0018
One cancellation, end to end	**2,843**	**551**	**$0.0042**	**$0.0084**

ONE CANCELLATION

$0.0042

of Gemini, handled end to end (2026 prices)

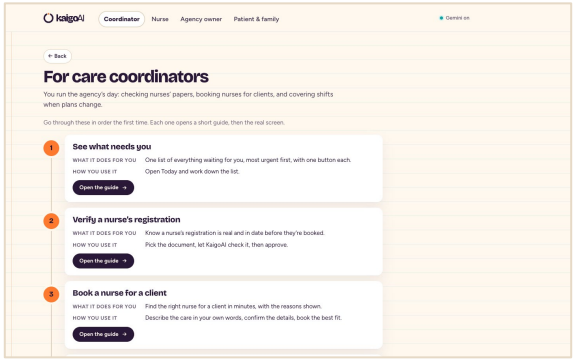
EXAMPLE AGENCY MONTH

$0.41

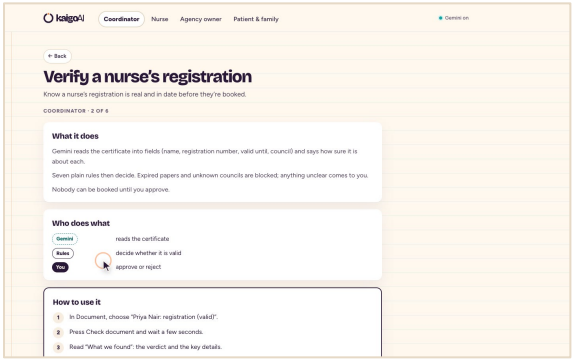
50 certificates, 60 requests, 40 cancellations (an assumption for illustration)

Infrastructure as deployed: Cloud Run 1 vCPU / 1 GiB, min 1 and max 5 instances, shared by every agency; Firestore, Cloud Storage, Maps Routes and Scheduler are pay-per-use at demo volumes.
Prices: Gemini 3.8 Flash on Vertex AI, standard: $0.75 / $3.75 per 1M input / output tokens to 31 Dec 2026; $1.50 / $7.50 from 1 Jan 2027 (cloud.google.com/vertex-ai/generative-ai/pricing). Token counts measured on the live app, 12 Oct 2026; they exclude thinking tokens (thinking level LOW), so budget about 2× for safety.

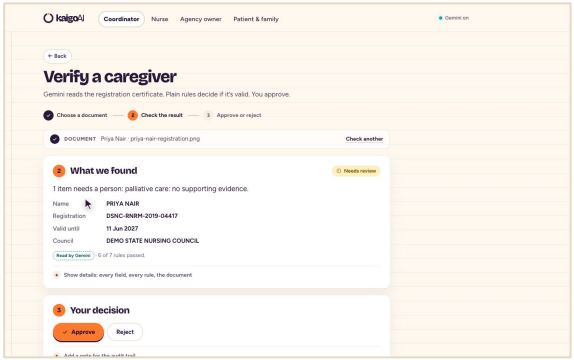
Snapshots of the prototype



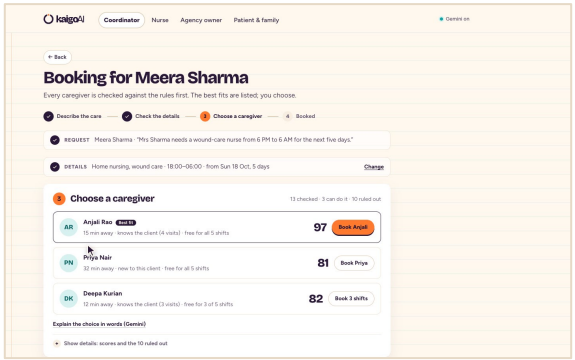
Coordinator tab: what KaigoAI does for you



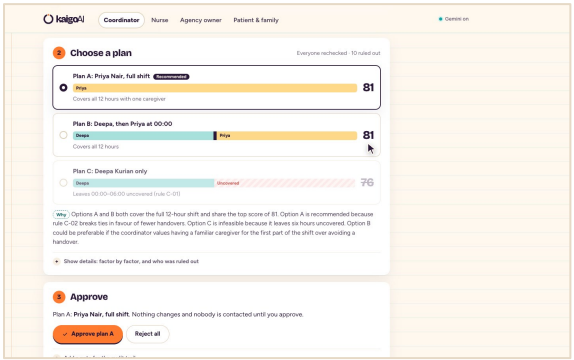
A guide: who does what, and how to use it



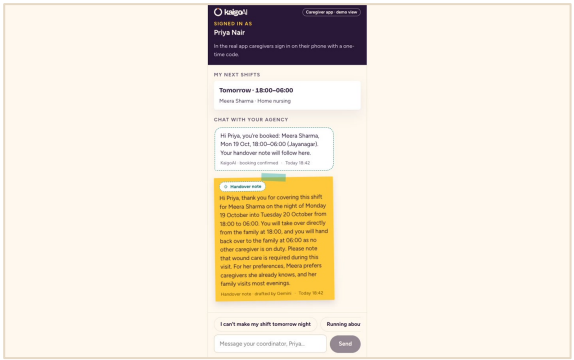
Verify: what we found, then approve



Book: choose a caregiver



Replan: choose a plan



Nurse's phone: booking and handover note

Prototype Performance report/Benchmarking

31/31
certificate fields read correctly

34/34
care-request checks passed

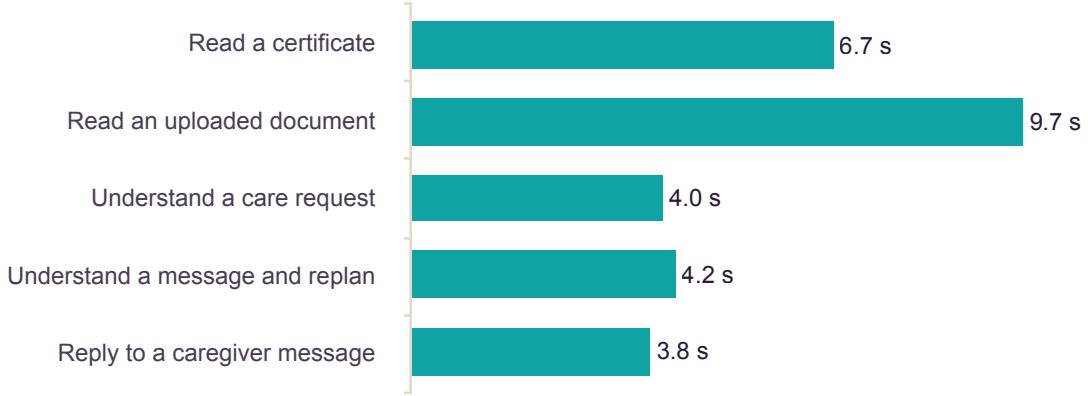
8/8
caregiver messages understood

3/3
explanations grounded

2–7 s
median Gemini call, by task

Every run of three on the synthetic evaluation set (docs/evaluation.md).

Live end-to-end time per AI-backed action (s), UAT 12 Oct 2026



UAT, LIVE APP, 12 OCT

84 of 84 cases passed

AUTOMATED BACKEND TESTS

61 passing (rules, workflows, AI retry, resets)

V6 SMOKE TEST, 18 OCT

74 of 74 page checks; 0 console errors

DEFECTS FOUND IN UAT

9 found, 8 fixed and re-tested, 1 open (low)

Additional Details/Future Development (if any)

BUSINESS CASE

Who pays
Home-nursing, hospice and elder-care agencies; the coordinator is the daily user.

Model
SaaS per agency, priced by active caregivers per month (hypothesis to validate in pilots).

Go to market
Pilot with 2–3 Bengaluru agencies, then city by city through agency networks.

Why we win
Continuity data and per-vertical rules improve with every replan; switching costs grow with the audit trail.

ROADMAP

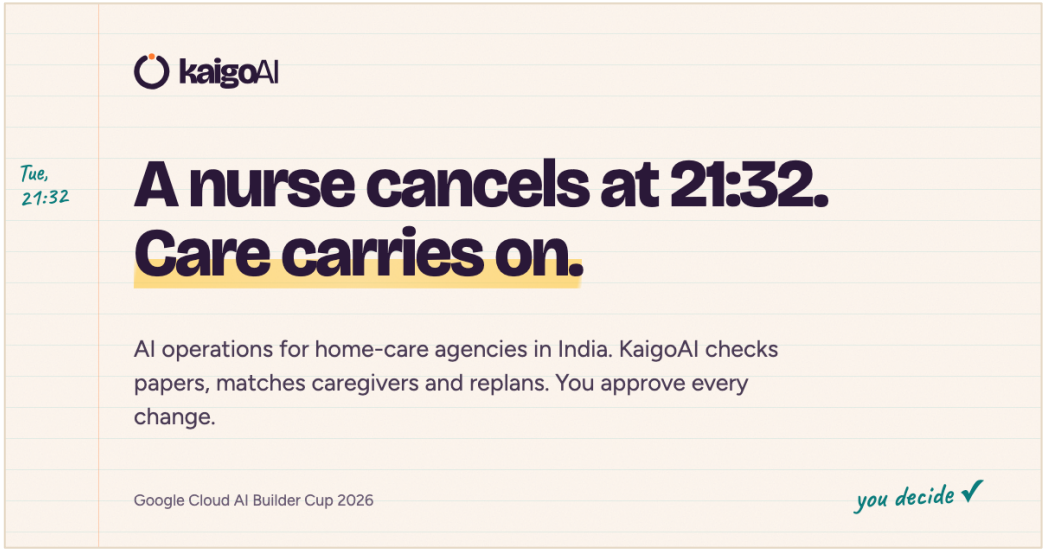
Next 3 months
Pilots measuring time to cover a cancellation, uncovered care hours and handovers per night; WhatsApp and SMS intake; phone OTP sign-in for caregivers.

Next 12 months
Elder care, disability support and childcare as configuration; Hindi and Kannada messages; payroll and HR integrations.

Guardrails we keep
People approve every change; rules stay code; nothing clinical; data minimised and private by design.

Provide links to your:

- GitHub Public Repository: https://github.com/data-with-saumya/kaigo-ai
- Demo Video Link (3 Minutes): https://youtu.be/Tr6l-tZZjfs
- Final Product Link: https://kaigoai-v6.web.app



Try it in two minutes

1. Open **kaigoai-v6.web.app** and press "Open the app".
2. Pick your tab: Coordinator, Nurse, Agency owner, or Patient & family.
3. Open a guide and press its button: it sets up the demo and opens the real screen. Back is on every page.

Every person and document is synthetic; each visitor gets a private demo agency.

Responsible AI and social impact

People decide
Every staffing change waits for a coordinator's approval.

Rules stay rules
Expiry, eligibility, availability, overlaps and coverage are code, never AI.

Nothing invented
Schema-checked AI; explanations citing unknown figures are withheld; unreadable values stay empty.

Not clinical
No diagnoses or treatment advice; medical details in requests are dropped.

WHO BENEFITS

People cared for at home and their families: care continues through cancellations, with a familiar nurse where possible

Caregivers: clear bookings, fewer midnight calls, handover notes before they arrive

Coordinators: one decision instead of an evening of phone calls

The community: expired or unverified caregivers are kept out of homes

We will measure in pilots
Time to cover a cancellation, uncovered care hours, handovers per night, credential problems caught before a shift.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

